

DSC 291: Learning Theory Assignment 3

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1 Part A: A Mini-Course on Concentration Inequalities

1.1 Notation and Setup

Throughout this mini-course I use one notation system across all four inequalities.

- $X_1, \dots, X_n$ are random variables.
- $\mu_i = \mathbb{E}[X_i]$ and, when all means are the same, $\mu = \mathbb{E}[X_i]$.
- The centered sum is

$$S_n = \sum_{i=1}^n (X_i - \mu_i).$$

- The empirical average is

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

- For Bernstein's inequality, the total variance proxy is

$$V_n = \sum_{i=1}^n \text{Var}(X_i).$$

- For bounded-difference inequalities, $f(X_1, \dots, X_n)$ is a function of independent inputs, and c_i denotes the sensitivity of f to the i th coordinate.

When I switch to learning-theoretic notation later, I will do so explicitly and then reduce back to these same concentration templates.

1.2 The Concept of Concentration

1.2.1 What Concentration Means

A concentration inequality is a statement of the form

$$\mathbb{P}(|Y - \mathbb{E}[Y]| \geq t) \leq \text{small tail bound in } t,$$

where Y is a random quantity built from many smaller random ingredients.

The point is not merely that Y has an expectation. The point is that Y is unlikely to wander far from that expectation. When the tail bound decays exponentially in t^2 or in t , the mean becomes a genuinely useful summary of the random variable.

1.2.2 Why This Should Happen

There are two recurring reasons concentration appears.

Averaging effect. If Y is a sum or average of many independent or weakly dependent terms, positive and negative fluctuations tend to cancel.

Stability effect. If $Y = f(X_1, \dots, X_n)$ and changing one input changes f only a little, then no single coordinate can move Y very far. This is the viewpoint behind McDiarmid's inequality.

1.2.3 Concrete Example

Let $X_1, \dots, X_n$ be i.i.d. Bernoulli(p) variables, so each $X_i \in \{0, 1\}$ and $\mathbb{E}[X_i] = p$. Then

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

is the sample frequency of success, while p is the true success probability.

We expect $\bar{X}_n$ to be close to p because each individual flip contains limited information and the average pools many such observations. Hoeffding's inequality makes that intuition quantitative:

$$\mathbb{P}(|\bar{X}_n - p| \geq \varepsilon) \leq 2e^{-2n\varepsilon^2}.$$

So the failure probability decays exponentially fast in n .

1.3 The Common Technique: MGF and Chernoff Bounds

1.3.1 The MGF Point of View

For a real-valued random variable Y , its moment generating function is

$$M_Y(\lambda) = \mathbb{E}[e^{\lambda Y}], \quad \lambda \in \mathbb{R}$$

whenever this expectation exists.

The MGF is useful because large values of Y are amplified exponentially by $e^{\lambda Y}$. If we can show that $\mathbb{E}[e^{\lambda Y}]$ is small enough, then large positive excursions of Y must be rare.

1.3.2 Chernoff Bound as the General Template

Lemma 1.1 (Chernoff bound template). *Let Y be any random variable for which $\mathbb{E}[e^{\lambda Y}] < \infty$ for some $\lambda > 0$. Then for every $t > 0$,*

$$\mathbb{P}(Y \geq t) \leq e^{-\lambda t} \mathbb{E}[e^{\lambda Y}].$$

Consequently,

$$\mathbb{P}(Y \geq t) \leq \inf_{\lambda > 0} e^{-\lambda t} \mathbb{E}[e^{\lambda Y}].$$

Proof. By Markov's inequality applied to the nonnegative random variable $e^{\lambda Y}$,

$$\mathbb{P}(Y \geq t) = \mathbb{P}(e^{\lambda Y} \geq e^{\lambda t}) \leq e^{-\lambda t} \mathbb{E}[e^{\lambda Y}].$$

Now optimize over $\lambda > 0$. □

So every MGF-based proof has the same skeleton:

1. choose the random quantity Y whose tail you want to control;
2. bound $\mathbb{E}[e^{\lambda Y}]$;
3. plug that bound into Lemma 1.1;
4. optimize in λ .

1.3.3 What Needs to Be Controlled

To get a useful tail bound, we want the centered MGF

$$\mathbb{E}[e^{\lambda(Y - \mathbb{E}[Y])}]$$

to look like the MGF of a well-understood distribution.

- If it is bounded by $e^{c\lambda^2}$, then we obtain Gaussian-type tails $e^{-t^2/(4c)}$.
- If it is bounded by something like $e^{\lambda^2 v / (2(1 - \lambda b/3))}$, then we obtain Bernstein-type tails, which use both a variance term v and a range bound b .

1.3.4 The Bounded-Differences Variant

For McDiarmid's inequality the random quantity is not usually a plain sum. Instead we write

$$f(X_1, \dots, X_n) - \mathbb{E}[f(X_1, \dots, X_n)]$$

as a martingale sum

$$\sum_{i=1}^n D_i$$

where each increment D_i has conditional mean zero and controlled range. Then we run the same MGF method conditionally, one increment at a time.

So the real common template is broader than “sum of independent variables.” It is:

Represent the fluctuation as a sum of increments with controlled exponential moments, and then apply Chernoff.

1.4 Theorem Statements, Assumptions, and Consequences

1.4.1 Hoeffding's Inequality

Theorem 1.2 (Hoeffding inequality [1]). *Let $X_1, \dots, X_n$ be independent random variables such that*

$$a_i \leq X_i \leq b_i \quad \text{almost surely for } i = 1, \dots, n.$$

Then for every $t > 0$,

$$\mathbb{P}\left(\sum_{i=1}^n (X_i - \mu_i) \geq t\right) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Equivalently, for every $\varepsilon > 0$,

$$\mathbb{P}\left(\bar{X}_n - \frac{1}{n} \sum_{i=1}^n \mu_i \geq \varepsilon\right) \leq \exp\left(-\frac{2n^2 \varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

The same bound holds for the lower tail, and therefore the two-sided bound is

$$\mathbb{P}\left(\left|\sum_{i=1}^n (X_i - \mu_i)\right| \geq t\right) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

This theorem controls sums of *independent bounded variables*. It does not use any variance information beyond boundedness, so it is robust but sometimes conservative.

1.4.2 Sampling Without Replacement

Theorem 1.3 (Hoeffding inequality for sampling without replacement [1, 2]). *Let $z_1, \dots, z_N \in [0, 1]$ be fixed numbers and let $Z_1, \dots, Z_n$ be sampled uniformly without replacement from $\{z_1, \dots, z_N\}$. Let*

$$\mu_N = \frac{1}{N} \sum_{i=1}^N z_i.$$

Then for every $\varepsilon > 0$,

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n Z_i - \mu_N\right| \geq \varepsilon\right) \leq 2e^{-2n\varepsilon^2}.$$

This inequality looks formally identical to the i.i.d. bounded case. The striking point is that sampling without replacement is *not* independent, yet the same sub-Gaussian rate survives. The key structural reason is negative dependence: once a large value is sampled, it becomes unavailable later.

1.4.3 McDiarmid's Inequality

Theorem 1.4 (McDiarmid inequality [3]). *Let $X_1, \dots, X_n$ be independent random variables taking values in sets $\mathcal{X}_1, \dots, \mathcal{X}_n$. Let*

$$f : \mathcal{X}_1 \times \dots \times \mathcal{X}_n \rightarrow \mathbb{R}$$

be such that for every i and every two inputs differing only in the i th coordinate,

$$|f(x_1, \dots, x_n) - f(x_1, \dots, x'_i, \dots, x_n)| \leq c_i.$$

Then for every $t > 0$,

$$\mathbb{P}(f(X_1, \dots, X_n) - \mathbb{E}[f(X_1, \dots, X_n)] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

The same holds for the lower tail, hence

$$\mathbb{P}(|f(X_1, \dots, X_n) - \mathbb{E}[f(X_1, \dots, X_n)]| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

This theorem controls *stable functions of independent inputs*. It is especially useful when f is not a linear statistic. In learning theory, it is the natural tool for bounding the fluctuation of a supremum such as a uniform deviation.

1.4.4 Bernstein's Inequality

Theorem 1.5 (Bernstein inequality for bounded summands [4]). *Let $X_1, \dots, X_n$ be independent random variables with*

$$\mathbb{E}[X_i] = 0 \quad \text{and} \quad |X_i| \leq b$$

almost surely for every i . Let

$$V_n = \sum_{i=1}^n \mathbb{E}[X_i^2].$$

Then for every $t > 0$,

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq t\right) \leq \exp\left(-\frac{t^2}{2(V_n + bt/3)}\right).$$

Hence

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i\right| \geq t\right) \leq 2 \exp\left(-\frac{t^2}{2(V_n + bt/3)}\right).$$

Bernstein improves on Hoeffding when the variance is much smaller than the worst-case range. For moderate deviations, the dominant scale is V_n rather than nb^2 .

1.4.5 Comparison of the Four Inequalities

- Theorem 1.2 assumes independence and bounded support. It gives a universal sub-Gaussian tail.
- Theorem 1.3 removes independence but exploits the special finite-population structure of sampling without replacement.
- Theorem 1.4 no longer asks for a sum. Instead it asks for bounded coordinate influence.
- Theorem 1.5 adds a variance term. When $V_n \ll nb^2$, it is much sharper than Hoeffding.

The main proof-level distinction is that Bernstein uses *more distributional information* than Hoeffding. Hoeffding only remembers the range of each summand; Bernstein remembers both the range and the variance.

1.5 Proofs

1.5.1 A Reusable MGF Lemma

Lemma 1.6 (Hoeffding lemma [1]). *If X is a random variable satisfying $a \leq X \leq b$ almost surely, then for every $\lambda \in \mathbb{R}$,*

$$\mathbb{E}[e^{\lambda(X - \mathbb{E}[X])}] \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

Proof. Let $m = \mathbb{E}[X]$ and write

$$p = \frac{m-a}{b-a} \in [0, 1].$$

By convexity of the exponential function, for every $x \in [a, b]$,

$$e^{\lambda x} \leq \frac{b-x}{b-a} e^{\lambda a} + \frac{x-a}{b-a} e^{\lambda b}.$$

Taking expectations yields

$$\mathbb{E}[e^{\lambda X}] \leq (1-p)e^{\lambda a} + pe^{\lambda b}.$$

Therefore

$$\mathbb{E}[e^{\lambda(X-m)}] \leq e^{-\lambda m}((1-p)e^{\lambda a} + pe^{\lambda b}).$$

Set $h = \lambda(b-a)$. Since $m = a + p(b-a)$, the right-hand side becomes

$$e^{-ph}(1-p+pe^h).$$

Define

$$g(h) = \log(1-p+pe^h) - ph.$$

Then $g(0) = 0$ and $g'(0) = 0$. A direct computation gives

$$g''(h) = \frac{pe^h(1-p)}{(1-p+pe^h)^2} \leq \frac{1}{4}.$$

Hence Taylor's theorem around 0 implies

$$g(h) \leq \frac{h^2}{8}.$$

Exponentiating,

$$\mathbb{E}[e^{\lambda(X-m)}] \leq e^{h^2/8} = \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

□

1.5.2 Proof of Hoeffding's Inequality

Proof of Theorem 1.2. Fix $\lambda > 0$. By independence,

$$\mathbb{E}[e^{\lambda S_n}] = \prod_{i=1}^n \mathbb{E}[e^{\lambda(X_i - \mu_i)}].$$

Applying Lemma 1.6 to each term,

$$\mathbb{E}[e^{\lambda S_n}] \leq \exp\left(\frac{\lambda^2}{8} \sum_{i=1}^n (b_i - a_i)^2\right).$$

Now Lemma 1.1 gives

$$\mathbb{P}(S_n \geq t) \leq \exp\left(-\lambda t + \frac{\lambda^2}{8} \sum_{i=1}^n (b_i - a_i)^2\right).$$

The right-hand side is minimized at

$$\lambda = \frac{4t}{\sum_{i=1}^n (b_i - a_i)^2}.$$

Substituting this value gives

$$\mathbb{P}(S_n \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

The lower-tail bound follows by applying the same argument to $-X_i$, and the two-sided bound follows by the union bound. $\square$

1.5.3 The Key Trick for Sampling Without Replacement

Theorem 1.7 (Hoeffding's convex comparison principle [1]). *Let $z_1, \dots, z_N$ be fixed real numbers. Let $Z_1, \dots, Z_n$ be sampled without replacement from this population, and let $W_1, \dots, W_n$ be sampled with replacement from the same population. Then for every convex function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$,*

$$\mathbb{E}\left[\varphi\left(\sum_{i=1}^n Z_i\right)\right] \leq \mathbb{E}\left[\varphi\left(\sum_{i=1}^n W_i\right)\right].$$

I am using Theorem 1.7 as the imported comparison theorem from Hoeffding's paper. It is the key proof trick for Theorem 1.3: without-replacement sampling is more concentrated than with-replacement sampling for every convex test function.

Proof of Theorem 1.3. Let

$$\mu_N = \frac{1}{N} \sum_{i=1}^N z_i.$$

Let $W_1, \dots, W_n$ be sampled with replacement from the same population. Then the W_i are i.i.d., each lies in $[0, 1]$, and each has mean μ_N .

Apply Theorem 1.7 with the convex function

$$\varphi(x) = e^{\lambda x}.$$

This gives, for every $\lambda > 0$,

$$\mathbb{E}\left[e^{\lambda \sum_{i=1}^n (Z_i - \mu_N)}\right] \leq \mathbb{E}\left[e^{\lambda \sum_{i=1}^n (W_i - \mu_N)}\right].$$

By Theorem 1.2 applied to the i.i.d. variables $W_i \in [0, 1]$,

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n W_i - \mu_N \geq \varepsilon\right) \leq e^{-2n\varepsilon^2}.$$

Since the without-replacement sample has no larger exponential moments than the with-replacement sample, the same Chernoff optimization gives

$$\mathbb{P}\left(\frac{1}{n} \sum_{i=1}^n Z_i - \mu_N \geq \varepsilon\right) \leq e^{-2n\varepsilon^2}.$$

Apply the same argument to $1 - Z_i$ to control the lower tail, then use the union bound to obtain

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n Z_i - \mu_N\right| \geq \varepsilon\right) \leq 2e^{-2n\varepsilon^2}.$$

$\square$

1.5.4 Proof of McDiarmid's Inequality

Proof of Theorem 1.4. Let

$$M_i = \mathbb{E}[f(X_1, \dots, X_n) \mid X_1, \dots, X_i], \quad i = 0, \dots, n,$$

with $M_0 = \mathbb{E}[f(X_1, \dots, X_n)]$ and $M_n = f(X_1, \dots, X_n)$. Then $(M_i)_{i=0}^n$ is the Doob martingale of f . Define the martingale differences

$$D_i = M_i - M_{i-1}.$$

Then

$$f(X_1, \dots, X_n) - \mathbb{E}[f(X_1, \dots, X_n)] = \sum_{i=1}^n D_i.$$

Fix i and condition on $X_1, \dots, X_{i-1}$. If we vary only the i th coordinate while averaging over the future coordinates $X_{i+1}, \dots, X_n$, the value of the conditional expectation can change by at most c_i . Therefore, conditional on $X_1, \dots, X_{i-1}$, the random variable D_i has mean zero and lies in an interval of length at most c_i .

So the conditional version of Lemma 1.6 gives

$$\mathbb{E}[e^{\lambda D_i} \mid X_1, \dots, X_{i-1}] \leq \exp\left(\frac{\lambda^2 c_i^2}{8}\right).$$

Iterating this bound,

$$\mathbb{E}\left[e^{\lambda \sum_{i=1}^n D_i}\right] \leq \exp\left(\frac{\lambda^2}{8} \sum_{i=1}^n c_i^2\right).$$

Applying Lemma 1.1,

$$\mathbb{P}\left(\sum_{i=1}^n D_i \geq t\right) \leq \exp\left(-\lambda t + \frac{\lambda^2}{8} \sum_{i=1}^n c_i^2\right).$$

Optimizing at

$$\lambda = \frac{4t}{\sum_{i=1}^n c_i^2}$$

gives

$$\mathbb{P}(f(X_1, \dots, X_n) - \mathbb{E}[f(X_1, \dots, X_n)] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

The lower-tail and two-sided bounds follow as before. □

1.5.5 A Scalar Inequality for Bernstein's Proof

Lemma 1.8. *For every real $u < 3$,*

$$e^u \leq 1 + u + \frac{u^2}{2(1 - u/3)}.$$

Proof. Consider

$$q(u) = \log\left(1 + u + \frac{u^2}{2(1 - u/3)}\right) - u.$$

A direct calculation shows $q(0) = 0$, $q'(0) = 0$, and $q'(u) \geq 0$ for $u < 3$. Hence $q(u) \geq 0$, which is equivalent to the claimed inequality. □

1.5.6 Proof of Bernstein's Inequality

Proof of Theorem 1.5. Fix $\lambda \in (0, 3/b)$. Since $|X_i| \leq b$, we have $\lambda X_i < 3$ almost surely. Applying Lemma 1.8 with $u = \lambda X_i$ and using $\mathbb{E}[X_i] = 0$,

$$\mathbb{E}[e^{\lambda X_i}] \leq 1 + \frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda b/3)}.$$

Using $1 + x \leq e^x$,

$$\mathbb{E}[e^{\lambda X_i}] \leq \exp\left(\frac{\lambda^2 \mathbb{E}[X_i^2]}{2(1 - \lambda b/3)}\right).$$

By independence,

$$\mathbb{E}\left[e^{\lambda \sum_{i=1}^n X_i}\right] \leq \exp\left(\frac{\lambda^2 V_n}{2(1 - \lambda b/3)}\right).$$

Chernoff's bound therefore yields

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq t\right) \leq \exp\left(-\lambda t + \frac{\lambda^2 V_n}{2(1 - \lambda b/3)}\right).$$

Now choose

$$\lambda = \frac{t}{V_n + bt/3}.$$

Then $\lambda < 3/b$, so this choice is admissible. A direct substitution gives

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq t\right) \leq \exp\left(-\frac{t^2}{2(V_n + bt/3)}\right).$$

Applying the same argument to $-X_i$ gives the lower tail and hence the two-sided version. $\square$

1.5.7 Proof Tricks Worth Reusing

- **Hoeffding:** reduce everything to an exponential-moment bound for a single bounded centered variable.
- **Sampling without replacement:** compare to a more convenient model. The whole theorem is driven by the convex comparison principle, Theorem 1.7.
- **McDiarmid:** convert a complicated function into a sum of martingale increments whose ranges are controlled one coordinate at a time.
- **Bernstein:** pay attention to the second moment. The variance term is the extra structure that buys a sharper tail.

1.6 Connection to the Week 3 ERM Proof

1.6.1 The Learning-Theoretic Goal

Let

$$\hat{h} \in \arg \min_{h \in \mathcal{H}} L_n(h)$$

be an empirical risk minimizer, where

$$L_n(h) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[h(x_i) \neq y_i]$$

and

$$L_{\mathcal{D}}(h) = \mathbb{P}_{(x,y) \sim \mathcal{D}}(h(x) \neq y).$$

The goal is to prove a high-probability bound of the form

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + O\left(\sqrt{\frac{\log \Gamma_{\mathcal{H}}(2n) + \log(1/\delta)}{n}}\right).$$

1.6.2 Step 1: A Uniform Deviation Quantity

Define

$$\Delta(S) = \sup_{h \in \mathcal{H}} |L_{\mathcal{D}}(h) - L_n(h)|.$$

If we can show that $\Delta(S)$ is small with high probability, then the usual ERM three-line comparison finishes the argument.

Lemma 1.9. *On the event $\{\Delta(S) \leq \varepsilon\}$,*

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + 2\varepsilon.$$

Proof. Let $h^* \in \arg \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h)$. Then

$$L_{\mathcal{D}}(\hat{h}) \leq L_n(\hat{h}) + \varepsilon \leq L_n(h^*) + \varepsilon \leq L_{\mathcal{D}}(h^*) + 2\varepsilon = \inf_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + 2\varepsilon.$$

□

1.6.3 Step 2: McDiarmid Controls the Fluctuation Around the Mean

Lemma 1.10. *For every $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $S \sim \mathcal{D}^n$,*

$$\Delta(S) \leq \mathbb{E}[\Delta(S)] + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

Proof. Changing one sample point can change each empirical risk $L_n(h)$ by at most $1/n$, while $L_{\mathcal{D}}(h)$ stays fixed. Therefore changing one coordinate of S changes $\Delta(S)$ by at most $1/n$. Applying Theorem 1.4 with $c_i = 1/n$ for all i gives

$$\mathbb{P}(\Delta(S) - \mathbb{E}[\Delta(S)] \geq t) \leq e^{-2nt^2}.$$

Setting

$$t = \sqrt{\frac{\log(1/\delta)}{2n}}$$

proves the claim. □

1.6.4 Step 3: Symmetrization

Let $S' \sim \mathcal{D}^n$ be an independent ghost sample, and let

$$L'_n(h) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}[h(x'_i) \neq y'_i].$$

Lemma 1.11 (Symmetrization).

$$\mathbb{E}[\Delta(S)] \leq \mathbb{E}_{S, S'} \left[\sup_{h \in \mathcal{H}} |L'_n(h) - L_n(h)| \right].$$

Proof. For each fixed h ,

$$L_{\mathcal{D}}(h) - L_n(h) = \mathbb{E}_{S'}[L'_n(h) - L_n(h)].$$

Therefore

$$\Delta(S) = \sup_{h \in \mathcal{H}} |\mathbb{E}_{S'}[L'_n(h) - L_n(h)]|.$$

By Jensen's inequality for the convex function $x \mapsto |x|$,

$$|\mathbb{E}_{S'}[L'_n(h) - L_n(h)]| \leq \mathbb{E}_{S'}[|L'_n(h) - L_n(h)|].$$

Taking the supremum over h and then using that $\sup \mathbb{E} \leq \mathbb{E} \sup$,

$$\Delta(S) \leq \mathbb{E}_{S'} \left[\sup_{h \in \mathcal{H}} |L'_n(h) - L_n(h)| \right].$$

Now take expectation over S . □

1.6.5 Step 4: Reduce to the Transductive Concentration Problem

Condition on the combined $2n$ sample points appearing in S and S' . After conditioning, the pair (S, S') is just a random split of a fixed pool of size $2n$ into two groups of size n each. For a fixed hypothesis h , define the pool labels

$$z_j(h) = \mathbf{1}[h(\tilde{x}_j) \neq \tilde{y}_j] \in \{0, 1\}, \quad j = 1, \dots, 2n,$$

where $(\tilde{x}_j, \tilde{y}_j)$ runs over the conditioned pool.

Then

$$L_n(h)$$

and

$$L'_n(h)$$

are the averages of the same $2n$ fixed numbers over a random split into two parts of size n .

Lemma 1.12. *For every $t > 0$ and every conditioned $2n$ -point pool,*

$$\mathbb{P}\left(\sup_{h \in \mathcal{H}} |L'_n(h) - L_n(h)| > t \mid S \cup S'\right) \leq 2\Gamma_{\mathcal{H}}(2n)e^{-nt^2/2}.$$

Proof. Fix h . Let

$$L_{\text{pool}}(h) = \frac{1}{2n} \sum_{j=1}^{2n} z_j(h).$$

Since the two halves have the same size,

$$L'_n(h) - L_n(h) = 2(L_{\text{pool}}(h) - L_n(h)).$$

By Theorem 1.3,

$$\mathbb{P}\left(|L_n(h) - L_{\text{pool}}(h)| > \frac{t}{2} \mid S \cup S'\right) \leq 2e^{-nt^2/2}.$$

Equivalently,

$$\mathbb{P}(|L'_n(h) - L_n(h)| > t \mid S \cup S') \leq 2e^{-nt^2/2}.$$

Now, on the conditioned pool, two hypotheses that induce the same binary labeling on the $2n$ points have exactly the same value of $|L'_n(h) - L_n(h)|$. So there are at most

$$\Gamma_{\mathcal{H}}(2n)$$

different quantities to consider. A union bound gives the claim. $\square$

1.6.6 Step 5: Tail Integration Bounds the Mean

Lemma 1.13.

$$\mathbb{E}[\Delta(S)] \leq 2\sqrt{\frac{2(\log(2\Gamma_{\mathcal{H}}(2n)) + 1)}{n}}.$$

Proof. By Lemma 1.11,

$$\mathbb{E}[\Delta(S)] \leq \mathbb{E}\left[\sup_{h \in \mathcal{H}} |L'_n(h) - L_n(h)|\right].$$

For any nonnegative random variable Y ,

$$\mathbb{E}[Y] = \int_0^\infty \mathbb{P}(Y > t) dt.$$

Applying this identity with

$$Y = \sup_{h \in \mathcal{H}} |L'_n(h) - L_n(h)|$$

and using Lemma 1.12 then removing the conditioning,

$$\mathbb{E}[Y] \leq \int_0^\infty \min\left\{1, 2\Gamma_{\mathcal{H}}(2n)e^{-nt^2/2}\right\} dt.$$

Set

$$a = \sqrt{\frac{2(\log(2\Gamma_{\mathcal{H}}(2n)) + 1)}{n}}.$$

Then

$$2\Gamma_{\mathcal{H}}(2n)e^{-na^2/2} = e^{-1}.$$

Hence

$$\mathbb{E}[Y] \leq a + \int_a^\infty 2\Gamma_{\mathcal{H}}(2n)e^{-nt^2/2} dt.$$

Using $t^2 - a^2 \geq 2a(t - a)$ for $t \geq a$,

$$\int_a^\infty 2\Gamma_{\mathcal{H}}(2n)e^{-nt^2/2} dt \leq 2\Gamma_{\mathcal{H}}(2n)e^{-na^2/2} \int_a^\infty e^{-na(t-a)} dt = \frac{1}{ena}.$$

Since $\log(2\Gamma_{\mathcal{H}}(2n)) + 1 \geq 1$, we have

$$\frac{1}{ena} \leq a.$$

Therefore

$$\mathbb{E}[Y] \leq 2a = 2\sqrt{\frac{2(\log(2\Gamma_{\mathcal{H}}(2n)) + 1)}{n}}.$$

□

1.6.7 Step 6: The Final ERM Guarantee

Theorem 1.14. *Let*

$$\hat{h} \in \arg \min_{h \in \mathcal{H}} L_n(h).$$

Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $S \sim \mathcal{D}^n$,

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + 4\sqrt{\frac{2(\log(2\Gamma_{\mathcal{H}}(2n)) + 1)}{n}} + 2\sqrt{\frac{\log(1/\delta)}{2n}}.$$

In particular,

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + O\left(\sqrt{\frac{\log \Gamma_{\mathcal{H}}(2n) + \log(1/\delta)}{n}}\right).$$

Proof. By Lemma 1.10 and Lemma 1.13, with probability at least $1 - \delta$,

$$\Delta(S) \leq 2\sqrt{\frac{2(\log(2\Gamma_{\mathcal{H}}(2n)) + 1)}{n}} + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

Apply Lemma 1.9 with this value of ε .

□

1.6.8 What Each Inequality Contributed

- Theorem 1.2 proves concentration for independent bounded averages.
- Theorem 1.3 is the transductive concentration step after conditioning on the $2n$ sampled points.
- Theorem 1.4 controls the fluctuation of the supremum $\Delta(S)$ around its mean.
- Theorem 1.5 is not needed for the Week 3 ERM theorem as stated, but it is the next natural refinement when variance information is available and one wants sharper rates than Hoeffding can provide.

1.7 References Used

References

- [1] Wassily Hoeffding. Probability Inequalities for Sums of Bounded Random Variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. Available via Taylor & Francis: <https://www.tandfonline.com/doi/abs/10.1080/01621459.1963.10500830>.
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- [3] Colin McDiarmid. On the Method of Bounded Differences. In *Surveys in Combinatorics*, pages 148–188. Cambridge University Press, 1989. Publisher page: <https://www.cambridge.org/core/books/surveys-in-combinatorics-1989/on-the-method-of-bounded-differences/BD71F4D1C287C6E7450A7E12D4C3E082>.
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2 Part B: The No-Free-Lunch Theorem and the Fundamental Theorem

2.1 Problem 1

2.1.1 Quantifiers and Adversarial Choices

The theorem quantifies over:

- every learning algorithm A ,
- every sample size n satisfying $n < |X|/2$.

After these are fixed, the adversary is allowed to choose:

- a set of $2n$ distinct domain points,
- a bad target labeling on those points,
- the realizable distribution supported on those labeled points.

The proof first averages over all labelings of a fixed $2n$ -point subset, and then uses the probabilistic method to extract one specific bad labeling.

2.1.2 The Averaging Construction

Fix any learning algorithm A and any n with $n < |X|/2$. Since $|X| > 2n$, we may choose distinct points

$$x_1, \dots, x_{2n} \in X.$$

For each labeling vector

$$b = (b_1, \dots, b_{2n}) \in \{0, 1\}^{2n},$$

define a target function $f_b : X \rightarrow \{0, 1\}$ by

$$f_b(x_i) = b_i \quad \text{for } i = 1, \dots, 2n,$$

and define it arbitrarily outside $\{x_1, \dots, x_{2n}\}$.

Now define a distribution D_b on $X \times \{0, 1\}$ by

$$D_b((x_i, b_i)) = \frac{1}{2n}, \quad i = 1, \dots, 2n.$$

Thus D_b is uniform on the $2n$ labeled points, and

$$L_{D_b}(f_b) = 0.$$

So every D_b is realizable.

2.1.3 Average Population Loss Over Random Labelings

Let $B = (B_1, \dots, B_{2n})$ be uniformly distributed on $\{0, 1\}^{2n}$. An equivalent sampling procedure for $S \sim D_B^n$ is:

1. sample B uniformly from $\{0, 1\}^{2n}$,
2. sample indices $J_1, \dots, J_n$ i.i.d. uniformly from $[2n]$, independently of B ,
3. output

$$S = ((x_{J_1}, B_{J_1}), \dots, (x_{J_n}, B_{J_n})).$$

Let

$$V(S) = \{i \in [2n] : x_i \text{ appears at least once in } S\}$$

be the set of observed points, and let

$$U(S) = [2n] \setminus V(S)$$

be the unseen points. Since the sample contains only n examples,

$$|V(S)| \leq n,$$

so

$$|U(S)| \geq n.$$

Now condition on the realized sample S . The hypothesis

$$\hat{h} = A(S)$$

is then fixed. For any $i \in U(S)$, the bit B_i was never revealed in the sample, so conditional on S it is still an unbiased fair coin, independent of $\hat{h}(x_i)$. Therefore

$$\mathbb{E}[\mathbf{1}[\hat{h}(x_i) \neq B_i] \mid S] = \frac{1}{2} \quad \text{for every } i \in U(S).$$

Hence

$$\mathbb{E}[L_{D_B}(A(S)) \mid S] = \frac{1}{2n} \sum_{i=1}^{2n} \mathbb{E}[\mathbf{1}[\hat{h}(x_i) \neq B_i] \mid S] \geq \frac{|U(S)|}{2n} \cdot \frac{1}{2} \geq \frac{1}{4}.$$

Taking expectation over S and B gives

$$\mathbb{E}_B \mathbb{E}_{S \sim D_B^n} [L_{D_B}(A(S))] \geq \frac{1}{4}.$$

2.1.4 Extracting a Single Bad Labeling

Since the average over all $b \in \{0, 1\}^{2n}$ is at least $1/4$, there exists at least one labeling b^* such that

$$\mathbb{E}_{S \sim D_{b^*}^n} [L_{D_{b^*}}(A(S))] \geq \frac{1}{4}.$$

Let

$$f^* = f_{b^*}, \quad D = D_{b^*}.$$

Then $L_D(f^*) = 0$, so the distribution is realizable.

2.1.5 From Expected Loss to Constant-Probability Failure

Let

$$Z = L_D(A(S)) \in [0, 1].$$

We already know

$$\mathbb{E}[Z] \geq \frac{1}{4}.$$

Set

$$p = \mathbb{P}(Z \geq 1/8).$$

Since $0 \leq Z \leq 1$, we have

$$\mathbb{E}[Z] \leq (1-p) \cdot \frac{1}{8} + p \cdot 1 = \frac{1}{8} + \frac{7p}{8}.$$

Combining this with $\mathbb{E}[Z] \geq 1/4$ yields

$$\frac{1}{4} \leq \frac{1}{8} + \frac{7p}{8},$$

so

$$p \geq \frac{1}{7}.$$

Therefore

$$\mathbb{P}_{S \sim D^n} (L_D(A(S)) \geq 1/8) \geq \frac{1}{7}.$$

This proves the No-Free-Lunch theorem.

2.2 Problem 2

2.2.1 Corollary Closing the Lower-Bound Direction

Corollary 2.1. *If $\text{VCdim}(\mathcal{H}) = \infty$, then $\mathcal{H}$ is not realizably PAC learnable. Consequently, $\mathcal{H}$ is not agnostically PAC learnable either.*

Proof. Suppose $\text{VCdim}(\mathcal{H}) = \infty$. Fix any learning algorithm A and any sample size n . Because the VC dimension is infinite, there exists a set

$$C = \{x_1, \dots, x_{2n}\} \subseteq X$$

that is shattered by $\mathcal{H}$. Apply the No-Free-Lunch theorem from Problem 1 to these $2n$ points. We obtain a distribution D supported on C and a labeling function

$$f^* : C \rightarrow \{0, 1\}$$

such that

$$\mathbb{P}_{S \sim D^n} (L_D(A(S)) \geq 1/8) \geq \frac{1}{7}.$$

Since C is shattered by $\mathcal{H}$, there exists some $h^* \in \mathcal{H}$ satisfying

$$h^*(x) = f^*(x) \quad \text{for all } x \in C.$$

Because D is supported on C , this implies

$$L_D(h^*) = 0.$$

So for every sample size n there exists a realizable distribution D on which algorithm A still has failure probability at least $1/7$ at target excess level $1/8$.

Taking

$$\varepsilon = \frac{1}{8}, \quad \delta = \frac{1}{7},$$

we see that no sample size can satisfy the realizable PAC requirement uniformly over all distributions. Hence $\mathcal{H}$ is not realizably PAC learnable.

Agnostic PAC learnability is stronger than realizable PAC learnability, so agnostic PAC learnability also fails. $\square$

2.2.2 Concrete Application to a Class of Infinite VC Dimension

Consider the class

$$\mathcal{H}_{\text{fin}} = \{x \mapsto \mathbf{1}[x \in F] : F \subseteq \mathbb{N} \text{ is finite}\}.$$

I claim that

$$\text{VCdim}(\mathcal{H}_{\text{fin}}) = \infty.$$

Indeed, let

$$C = \{x_1, \dots, x_m\} \subseteq \mathbb{N}$$

be any finite set. For any labeling

$$y_1, \dots, y_m \in \{0, 1\},$$

define

$$F = \{x_i : y_i = 1\}.$$

Then F is finite, so the hypothesis

$$h_F(x) = \mathbf{1}[x \in F]$$

belongs to $\mathcal{H}_{\text{fin}}$, and it satisfies

$$h_F(x_i) = y_i \quad \text{for all } i = 1, \dots, m.$$

Thus every finite set is shattered, so the VC dimension is infinite.

Applying Corollary 2.1, $\mathcal{H}_{\text{fin}}$ is not PAC learnable. More explicitly: for every n and every learning algorithm A , there exists a realizable distribution D_n supported on $\{1, \dots, 2n\}$ and a target hypothesis $h_n^* \in \mathcal{H}_{\text{fin}}$ such that

$$\mathbb{P}_{S \sim D_n^n} (L_{D_n}(A(S)) \geq 1/8) \geq \frac{1}{7}.$$

So this class gives a concrete example of the lower-bound direction of the Fundamental Theorem.

2.3 Problem 3

2.3.1 A Concrete Learning Rule

Fix a sample size n and domain

$$X = \{1, 2, \dots, 2n\}.$$

Define the learning rule A_{mem} as follows: given a training sample

$$S = ((x_1, y_1), \dots, (x_n, y_n)),$$

output the classifier

$$A_{\text{mem}}(S)(x) = \mathbf{1}[x \text{ appears in } S \text{ with label } 1].$$

So the learner predicts 1 exactly on points it has explicitly seen labeled 1, and predicts 0 everywhere else.

2.3.2 Bad Distribution and Target

Let the target function be

$$f^*(x) = 1 \quad \text{for all } x \in X.$$

Let D be the uniform distribution on

$$\{(1, 1), (2, 1), \dots, (2n, 1)\}.$$

Then

$$L_D(f^*) = 0.$$

Moreover, f^* belongs to $\mathcal{H}_{\text{fin}}$, since it is the indicator of the finite set $\{1, \dots, 2n\}$.

2.3.3 Failure Calculation

Given any sample $S \sim D^n$, let

$$V(S) = \{x \in X : x \text{ appears at least once in } S\}.$$

Since the sample has only n examples,

$$|V(S)| \leq n.$$

Therefore the unseen set

$$U(S) = X \setminus V(S)$$

satisfies

$$|U(S)| \geq n.$$

For every $x \in U(S)$, the rule $A_{\text{mem}}(S)$ predicts 0, while the target label is $f^*(x) = 1$. Hence

$$L_D(A_{\text{mem}}(S)) = \frac{|U(S)|}{2n} \geq \frac{n}{2n} = \frac{1}{2}.$$

So the learner fails not merely with constant probability, but deterministically:

$$\mathbb{P}_{S \sim D^n}(L_D(A_{\text{mem}}(S)) \geq 1/2) = 1.$$

In particular,

$$\mathbb{P}_{S \sim D^n}(L_D(A_{\text{mem}}(S)) \geq 1/8) = 1.$$

This is an explicit witness to the kind of failure that the abstract NFL theorem guarantees existentially.

2.4 Problem 4

2.4.1 Learning Setting

The Week 1 theorem lives in the *online transductive* mistake-bound setting: the learner receives one point at a time and is judged by the total number of mistakes on a finite domain.

The Week 3 theorem lives in the *batch i.i.d. PAC* setting: the learner receives an i.i.d. training sample and is judged by the population loss under an unknown distribution.

2.4.2 Adversary Type

In Week 1, the proof is driven by an *adaptive adversary*. At each round, the adversary can look at the learner's current prediction on a fresh point and then choose the opposite label. After the interaction ends, this adaptive construction can be repackaged as a single bad function f and a fixed enumeration of the domain.

In Week 3, the adversary is *oblivious*. It chooses a bad distribution D and a realizable target function f^* before the sample is drawn. The learner is then evaluated on the randomness of the i.i.d. sample, not on an adaptively chosen online sequence.

2.4.3 Form of the Conclusion

The Week 1 conclusion is deterministic and combinatorial:

there exists a sequence on which the learner makes $|X|$ mistakes.

It is a worst-case online impossibility result.

The Week 3 conclusion is probabilistic and statistical:

$$\mathbb{P}_{S \sim D^n} (L_D(A(S)) \geq 1/8) \geq \frac{1}{7}.$$

It is a lower bound on generalization performance under random sampling.

2.4.4 Why Both Are Called No-Free-Lunch

Both theorems say the same qualitative thing: without structural restrictions on the target class, no learner can succeed uniformly over all problems.

Week 1 says there is no universally low-mistake online learner on an unrestricted finite domain.

Week 3 says there is no universally PAC learner on an unrestricted class of labelings, because for every sample size there is a realizable distribution on which the learner still has constant population error with constant probability.

So the two theorems live in different models and produce quantitatively different lower bounds, but they both express the same underlying message: without a complexity restriction, learning has no free lunch.

3 Part C: AI Usage Report

1. Parts of the Assignment Where I Used AI

This time it keeps the same, but here I added some new skills and help unify the notations. See details as next sections.

2. Concrete AI Suggestions I Accepted

One suggestion I accepted was to separate each concentration inequality into the same template: theorem statement, assumptions, proof sketch, full proof, and proof trick.

Another suggestion I accepted was to present the i.i.d. ERM argument through the quantity

$$\Delta(S) = \sup_{h \in \mathcal{H}} |L_{\mathcal{D}}(h) - L_n(h)|$$

and then split the proof into McDiarmid for fluctuation, symmetrization for the mean, and transductive concentration after conditioning on $S \cup S'$.

3. Concrete AI Suggestions I Rejected or Substantially Modified

I rejected an early proof for the without-replacement Hoeffding inequality that tried to treat the sample as if it were independent.

4. How I Verified Correctness

For theorem statements in Part A, I checked the formulas and assumptions against the cited references and against the Week 3 slides. Also this part is covered in the previous course, so it's easy to check.

For the ERM proof, I checked each reduction separately: McDiarmid for the fluctuation term, symmetrization for replacing population risk by ghost-sample empirical risk, and the conditional transductive argument for the growth-function term $\Gamma_{\mathcal{H}}(2n)$.

5. The 5 Most Recent AI Workflow Changes and Why

(i) **Added the ‘theorem-ref’ skill.** This came directly from this assignment. I needed a workflow for theorem-heavy writing that forces theorem statements, labels, source citations, unified notation, proof sketches, full proofs, and proof-trick summaries.

(ii) **Added a citation-first rule for theorem sections.** For concentration inequalities, I now want the source attached at the theorem statement stage instead of added at the end. This reduces the chance of drifting into uncited folklore formulations.

(iii) **Added a unified-notation rule for theorem collections.** In earlier drafts it was too easy for different theorems to use unrelated symbols. For this assignment I standardized notation for means, centered sums, empirical averages, variance terms, and bounded-difference constants.

(iv) **Added PDF-to-statement extraction with ‘pypdf’.** Instead of repeatedly rereading the raw PDF, I now extract the assignment into a reusable companion ‘.tex’ file first. That made it easier to preserve the exact prompt and reuse it across turns.

(v) **Added a stronger compile-and-check loop for theory writeups.**